

Deep Brain Stimulation Signal Classification using Deep Belief Networks

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Abstract—An approach to modeling complex real-world data such as biomedical signals is to develop pattern recognition techniques and robust features that capture the relevant information. In this paper, we use a deep belief network (DBN) to predict subcortical structures of patients with Parkinson's disease based on microelectrode records (MER) obtained during deep brain stimulation (DBS). We report on experiments using a data set involving 52 MER for the structures: zona incerta (Zi), subthalamic nucleus (STN), thalamus nucleus (TAL), and substantia nigra (SNR). The results show that our chosen features and network architecture produces a 99.5% accuracy of detection and classification of the subcortical structures under study. Based on the results we conclude that deep belief networks could be used to predict subcortical structure—mainly the STN for neurostimulation.

I. INTRODUCTION

Parkinson's disease (PD) is thought to affect at least 100 in every 100000 people. The cardinal symptoms of tremor, bradykinesia, postural instability, and rigor result in substantial disability for patients with PD. During the course of the disease, up to 50% of patients will have symptoms refractory to medication and will experience drug-induced dyskinesias. Over activity of the globus pallidus internus (GPi) and the subthalamic nucleus (STN) is believed to be part of the pathophysiologic mechanism of PD. Parkinson's disease (PD) is a chronic progressive neurodegenerative disorder affecting multiple brain circuits leading to motor symptoms such as bradykinesia, rigidity, resting tremor, and loss of postural reflexes[1]. PD also has non-motor manifestations such as neuropsychiatric symptoms, cognitive abnormalities, autonomic disorders, and sleep [2].

A. Motivation for Deep Learning

Human information processing mechanisms (e.g. vision and speech) suggest the potential effectiveness of deep architectures for extracting complex structure and building internal representations from rich sensory inputs. For example, human speech production and perception systems are both equipped with clearly layered hierarchical structures in transforming information from the waveform level to the linguistic level and vice versa. Thus, it is natural to infer that the state of the art can be advanced in processing these types of media signals if efficient and effective deep learning algorithms

can be developed. Biomedical signal processing systems with deep architectures are composed of many layers of nonlinear processing stages, where each lower layer's outputs are fed to its immediate higher layer as the input.

The concept of deep learning originated from artificial neural network research. A multilayer perceptron with many hidden layers is a good example of the models with deep architectures [3]. Deep learning techniques have been applied to a wide variety of problems in recent years [4], [5], [6]. In many of these applications, algorithms based on deep learning have surpassed the previous state-of-art performance. At the heart of all deep learning algorithms is the domain independent idea of using hierarchical layers of learned abstraction to efficiently accomplish high-level task.

B. Deep Brain Stimulation

Parkinson's Disease (PD) is primarily related to substantia nigra degeneration and, thus, dopamine insufficiency. L-DOPA as a precursor of dopamine is the standard medication in PD. However, disease progression causes L-DOPA therapy efficiency decay (on-off symptom fluctuation), and neurologists often decide to classify patients for DBS (Deep Brain Stimulation) surgery.

Deep brain stimulation (DBS) [7] is considered a safe and well tolerated surgical procedure to alleviate Parkinson's disease and other movement disorders symptoms along with some psychiatric conditions. Over the last few decades DBS has been shown to provide remarkable therapeutic effect on carefully selected patients. DBS improves motor functions and therefore quality of life. To date, one main target has emerged in PD patients: the subthalamic nucleus.

DBS involves the surgical implantation of electrodes into deep structures of the brain to modulate brain circuitry in an effort to restore normal physiological function.

DBS has been used effectively for the treatment of movement disorders, including Parkinson's disease (PD), Essential tremor (ET), and dystonia, as well as for psychiatric disorders such as obsessive-compulsive disorder (OCD). In addition, DBS may exert its influence via the correction of aberrant neuronal activity. For example, in the setting of DBS for the treatment of PD, the loss of dopamine, which is known to be largely responsible for the pathophysiology of the disease,

results in changes in the underlying activity of cells within the basal ganglia[8].

The exact placement of the stimulator is fundamental for the sensory and motor effects specific to the subthalamus, since small deviations can affect adjacent structures and generate side effects[9]. Technological development has allowed a higher display resolution for the imaging processing of these types of structures[10].

Also, the development of recording systems for the spontaneous or induced electrical activity of such structures, allows defining their position limits more accurately, anatomic relationships with adjacent structures and behavior in relation to the movement or symptoms of Parkinson components, which turns in a suitable situation in order to safely establish and characterize the affected area. The risk of errors in the localization of the surgical target for deep brain electrical stimulation (DBS) requires the use of some form of intraoperative neurophysiological monitoring to confirm the correct destination during surgery. The purpose of the development and implementation of techniques of classification for the processing microelectrode records (MER) is of great importance today, as it allows the surgical team to determine the optimal location of the lesion or DBS [11], [12].

In this paper we calculate 6 features each of which measure different signal characteristics. These features quantify neuronal activity inherent in the subcortical structures, and allows their localization for greater accuracy during DBS. We then apply a deep belief network for supervised classification and show that it can identify and predict any of the subcortical structures with high precision. The 4 structures are:

- Thalamus (TAL)
- zona incerta (Zi)
- subthalamic nucleus (STN)
- substantia nigra (SNR)

II. DEEP LEARNING

Deep learning provides a means of a richer representation of data which allows for higher recognition rates at the cost of computational complexity. Deep networks consist of an input layer, multiple hidden layers that are invisible to the user and an output layer. This allows discovery of intricate structure within the data. Backpropagation is usually the training algorithm of choice due to its simplicity and ability to compute the representation in each layer from the representation in the previous layer[3].

A. Restricted Boltzman Machines

A Restricted Boltzman Machine (RBM) is a learning machine consisting of an input layer, sometimes called a visible layer, and a hidden layer. Each of these layers consists of one or more neurons and the neurons in each layer are connected by weights. Processing information is as follows: each neuron in the input layer takes a sample from dataset, multiplies it by a weight and adds a bias value before passing it on to the hidden layer. The hidden layer then transforms this data by applying an activation function. Considering all neurons in

the input and hidden layers we can express these operations mathematically as $\mathbf{h} = f(\mathbf{W}\mathbf{v} + \mathbf{b})$. Here $\mathbf{W}$ is a weight matrix which expresses the weights between each neuron in the input layer and each neuron in the hidden layer, $\mathbf{v}$ is a vector of values in the visible layer. This is typically will be training, testing or validation data. The vector $\mathbf{b}$ contains the biases for each input layer neuron. The vector $\mathbf{h}$ contains values in the hidden layer. The activation function $f(x)$ has a variety of possibilities, but is usually chosen as the sigmoid function

$$\frac{1}{1 + e^{-x}} \quad (1)$$

The RBM learns the conditional density of the output given the input and the weights, $p(\mathbf{h}|\mathbf{v}; \mathbf{W})$. Training an RBM with data allows it to approximate this conditional density. It does this by optimizing the following energy function

$$E(v, h) = - \sum_{i,j} v_i h_j w_{ij} - \sum_i b_i v_i - \sum_u b_u h_u \quad (2)$$

Several RBMs can be stacked for deeper representation with the restriction each input layer is connected to only one hidden layer.

B. Deep Belief Networks

Deep Belief Networks can be formed by stacking numerous RBMs and the last layer containing the labels. Fig. 1 shows the structure of a deep belief network network, the direction of processing and the conditional probabilities learned.

III. MATERIALS AND METHODS

A. Dataset

Intra-operative microelectrode records were acquired in 5 (4 males and 1 female) awake and unmedicated Parkinson's patients who were subject to deep brain implantation under electrostimulation. The patients, aged between 49 and 61 years, participated voluntarily assuming previously signed consent. Microelectrode records were made using the ISIS MER Inomed system, which is used to obtain an optimal location of the destination (target) through deep brain stimulation. Visualization of neural data started 10mm on the target data. Each 1mm a new location was created if the distance between the microelectrode and the destination point was greater than 3mm. For distances less than 3mm, the locations were created every 0.5mm.

Specialists in neurosurgery and neurophysiology labeled the obtained signals using the MER system. The acquisition time for each record was 2 seconds with a sampling frequency of 24kHz (24000 samples per second). In total, the database is comprised of 52 micro recordings, 13 for each of the subcortical structures: Thalamus nucleus (TAL), Zona Incerta (Zi), Subthalamic nucleus (STN) and Substantia Nigra (SNR). These surgical procedures were performed at the Institute of Epilepsy and Parkinson in Pereira, Colombia.

B. Features

We calculated a subset of the features mentioned in [10], to extract the information contained in each of the MER for the different subcortical structures under study. In the features below x_i corresponds to a sample in the dataset $\mathbf{X} = (x_1, x_2, \dots, x_N)$.

Curve length This feature is useful in predicting the stability of the values of a signal. If, in a given interval, the value of this feature is low, it is an indicative that the signal is stable, otherwise the signal is unstable. We calculate this feature using equation (3).

$$L = \sum_{i=1}^{N-1} |x_{i+1} - x_i| \quad (3)$$

Threshold The threshold of a signal is based on the deviation of the data. It provides a measure of how scattered the signal is in given window of size N and is calculated as follows:

$$\gamma = \frac{3}{N-1} \sqrt{\sum_{i=1}^N (x_i - \bar{X})^2} \quad (4)$$

where $\bar{X}$ is the mean of the dataset.

Peaks The number of peaks in a given signal is determined by:

$$\kappa = \frac{1}{2} \sum_{i=1}^{N-2} \max \{0, \text{sgn}(x_{i+1} - x_i) - \text{sgn}(x_{i+2} - x_{i+1})\} \quad (5)$$

where

$$\max(a, b) = \begin{cases} a & \text{if } a > b \\ b & \text{if } a < b \\ a \text{ or } b & \text{if } a = b \end{cases}$$

and

$$\text{sgn}(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -1 & \text{if } x < 0 \end{cases}$$

Root mean square This quantity is defined as the square root of the mean of the squares of the values of the signal. The root mean square (also known as quadratic mean) is determined by:

$$\delta = \sqrt{\frac{\sum_{i=1}^N x_i^2}{N}} \quad (6)$$

Average nonlinear energy The average nonlinear energy can be computed as:

$$ANE = \frac{1}{N-2} \sum_{i=2}^{N-1} x_i^2 - x_{i-1}x_{i+1} \quad (7)$$

Zero crossings The number of zero crossings κ for a given signal is determined through the formula:

$$\kappa = \frac{1}{2} \sum_{i=1}^{N-1} |\text{sgn } x_{i+1} - \text{sgn } x_i| \quad (8)$$

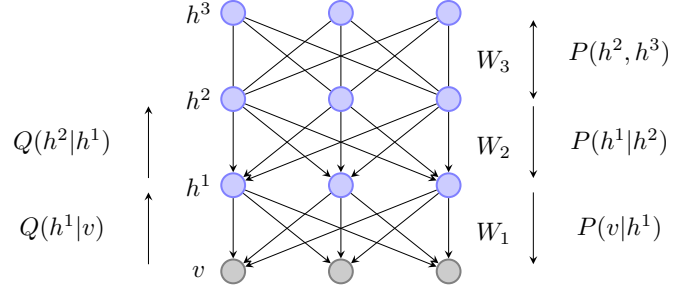


Fig. 1. A Deep Belief Network (DBN)

C. Tools

The features described in the previous section were calculated using Matlab R2015. We used the Python modules *nolearn* and *scikit-learn*[13] implementations for the Deep Belief Network.

D. Methodology

As previously mentioned, each record was acquired for 2 seconds at a sampling frequency of 24kHz. This leads to each record having 48000 samples. If we consider a trajectory of 13 records for each of the subcortical structures: Thalamus nucleus (TAL), Zona Incerta (Zi), Subthalamic nucleus (STN) and Substantia Nigra (SNR), the final trajectory is made up of 52 records and has a total of 2496000 samples. Next, the final trajectory is divided into windows 4992 consecutive samples and for each of these windows the six features are determined, yielding a total of 500 instances (patterns) by feature.

Our dataset consists of 125 instances from each of the 4 classes: Thalamus, Zona Incerta, Subthalamic nucleus, and Substantia Nigra. We train a DBN whose architecture is shown in Table I on a set of randomly selected patterns, approximately 60% while the remaining 40% is used as the testing set.

TABLE I
DEEP BELIEF NETWORK ARCHITECTURE

Variables	Parameters
Number of Input layer neurons	6
Number of Hidden layers	300
Number of Output layer neurons	4
Number of training epochs	100

IV. RESULTS

Tables II and III summarize the results of our experiments and are presented below. The results show a classification accuracy of 99.5% on testing data and near perfect precision. We also show the classifier's confusion matrix. The classification error, shown in Table III, was the result of Substantia Nigra (SNR) being incorrectly classified as Subthalamic nucleus.

TABLE II
DEEP BELIEF NETWORK TESTING RESULTS

Class	Precision	Recall	F-score	Support
TAL	1.0	1.0	1.0	61
Zi	1.0	1.0	1.0	44
STN	0.98	1.0	0.99	45
SNR	1.0	0.98	0.99	50
avg/total	1.0	0.99	1.0	200

TABLE III
CONFUSION MATRIX FOR DBN TESTING

TAL	Zi	STN	SNR
61	0	0	0
0	44	0	0
0	0	45	0
0	0	1	49

V. CONCLUSIONS

In this paper we investigated the use of deep belief networks for the classification and prediction of subcortical structures of patients with Parkinson's disease, based on microelectrode records (MER), obtained during deep brain stimulation (DBS).

We chose six types of input features and a deep belief network architecture to precisely predict subcortical structures. We demonstrated that the deep belief network has excellent classification properties with a relatively small amount of data.

One can infer that a deep learning approach is an appropriate analysis mechanism. Further these investigations could be improved in future studies by carrying out more exhaustive searches for the parameters in the architectures.

We conclude that we can produce a trained model that could be used to monitor the location of subcortical structures in real time, reducing the uncertainty that can exist during surgery, and representing a valuable support tool for neurosurgeons and electrophysiologists during electrical stimulation and deep brain electrode implantation for the treatment of Parkinson's patients.

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